

Pharmacological Characteristics of Ryokan-kyomi-shinge-nin-to, an Antiallergic Kampo Medicine

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The pharmacological characteristics of Ryokan-kyomi-shinge-nin-to (RKS), a traditional oriental herbal (Kampo) medicine which has been used for the treatment of allergic asthma and rhinitis, were investigated. The number of sneezes by actively sensitized mice after a topical antigen challenge was significantly reduced by pre-treatment with RKS (300 and 1000 mg/kg, *p.o.*). Although RKS did not inhibit the antigen-induced histamine release from rat peritoneal exudate cells (PEC), it significantly inhibited an increase in vascular permeability induced by histamine and serotonin. These results suggest that RKS has antiallergic activity in animals, and the functional antagonism of a histamine response may be one of the mechanisms of its effect. In addition, RKS prevented histamine hypersensitivity in actively sensitized mice. Because RKS did not affect sleeping time induced by pentobarbital in mice and did not inhibit gastric emptying in rats, the drug appears to be useful for treating allergic patients suffering from classical antihistamines side effects such as stomach discomfort or relative drowsiness.

Key words Ryokan-kyomi-shinge-nin-to; antiallergy; stomachic; ketotifen

Allergic asthma and rhinitis are the most common forms of atopic disease. However, many of the antiallergics used to treat these illnesses can cause gastrointestinal disorders.^{1,2)} The traditional herbal (Kampo) medicine Ryokan-kyomi-shinge-nin-to (RKS) has long been used for the treatment of allergic diseases in Japan where Kampo medicines are manufactured with a standardized quality and quantity of ingredients. However, few experimental data are available on the antiallergic properties of RKS, except for reports about studies on the release of histamine from mast cells.^{3,4)} In the present study, we investigated RKS for possible antiallergic effects in experimental models, with a special emphasis on possible side effects.

MATERIALS AND METHODS

Animals Male BALB/c mice (5 weeks old), male ICR mice (5 weeks old) and male SD (IGS) rats (5 weeks old) were purchased from Charles River, Inc. (Tsukuba, Japan), and housed at constant temperature ($23 \pm 2^\circ\text{C}$) and humidity ($55 \pm 10\%$) until use. All experiments were conducted according to the institution's guidelines for care and use of laboratory animals in research.

Drugs Ryokan-kyomi-shinge-nin-to (RKS; Tsumura & Co., Tokyo, Japan) is a powdered extract from a mixture of 7 medicinal plants: Hoelen (4 g), Glycyrrhizae Radix (2 g), Zingiberis Siccata Rhizoma (2 g), Schizandrae Fructus (3 g), Asiasari Radix (2 g), Pinelliae Tuber (4 g) and Armeniacae Semen (4 g). RKS, seratrodist (Takeda, Osaka, Japan), ketotifen fumarate and metoclopramide monohydrochloride (Sigma Chemical, St. Louis, MO, U.S.A.) were administered *p.o.* The other drugs were ovalbumin (OA; Fraction V), anti-dinitrophenyl (DNP) IgE, histamine dihydrochloride, serotonin creatinine sulfate complex, Evans blue (Sigma Chemical), aluminum hydroxide ($\text{Al}(\text{OH})_3$; Wako Pure Chemical Industries, Ltd., Osaka, Japan), DNP-BSA (SLS, Tokyo, Japan) and sodium pentobarbital (Dinabot Laboratoris, North Chicago, IL, U.S.A.).

Antigen-Induced Sneezing in Mice Male BALB/c mice were actively sensitized with an intraperitoneal injection of $1\ \mu\text{g}$ of OA containing 0.5 mg of $\text{Al}(\text{OH})_3$ in 0.2 ml of saline at 0, 1, 2 and 3 weeks. One week after the last sensitization, $20\ \mu\text{l}$ of OA in saline (40 mg/ml) was administered intranasally for 10 d. The next day, a topical antigen challenge was performed by dropping $20\ \mu\text{l}$ of OA solution (40 mg/ml) intranasally, and the number of sneezes was counted for 10 min. Mice were pretreated with 300 or 1000 mg/kg of RKS, 5 mg/kg of ketotifen or distilled water (solvent) 30 min before the antigen challenge.

Antigen-Induced Histamine Release from Rat Peritoneal Exudate Cells This test was conducted according to Abe *et al.*⁵⁾ In brief, male SD rats were administered 1 ml of monoclonal anti-DNP IgE (titer: 1000; i.p.). At 24 h, the animals were bled to death, and their bodies were injected i.p. with 10 ml of Tyrode's solution (pH 7.4) containing 0.3% BSA (Sigma Chemical) and 5 units/ml of heparin (Mochida Pharmaceutical, Tokyo, Japan). After gently massaging the abdomen for 2 to 3 min, peritoneal cells were obtained and washed 3 times with Tyrode's solution. The mast cells in a cell suspension were counted by staining with 0.05% toluidine blue (Kanto Chemicals), resuspended in Tyrode's solution containing 0.1% BSA and 10 mM HEPES buffer (Sigma Chemical), and adjusted to a concentration of 1×10^5 cells/ml. The cell suspension was preheated for 5 min to 37°C , and incubated with the test drugs for 10 min, challenged with DNP-BSA at a final concentration of $10\ \mu\text{g}/\text{ml}$ in the presence of $10\ \mu\text{g}/\text{ml}$ of phosphatidyl serine (Sigma Chemical) and incubated for an additional 10 min. The concentration of histamine in the supernatant was determined according to the method of Shore *et al.*⁶⁾ The percentage of histamine released was calculated from the total cellular content of histamine in each experiment.

Histamine- and Serotonin-Induced Skin Reactions in Rats This test was conducted according to Inagaki *et al.*⁷⁾ In brief, histamine ($1\ \mu\text{g}/0.1\ \text{ml}$), serotonin ($30\ \text{ng}/0.1\ \text{ml}$) and saline (solvent; $0.1\ \text{ml}$) were administered i.d. at 3 sites on

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the shaved backs of rats, and 3 different reactions were elicited in the same animal. Saline containing 1% Evans blue was immediately administered i.v. After 30 min, the rats were bled to death, the blue area on the dorsal skin was cut out, and Evans blue was extracted for measurement of absorbance at 620 nm. Test drugs were administered *p.o.* 1 h prior to the injection of the mediator.

Histamine-Hypersensitivity in Mice This test was conducted according to Watanabe *et al.*⁸⁾ Male BALB/c mice were actively sensitized with an intraperitoneal injection of 1 μ g of OA containing 0.1 mg of Al(OH)₃ in 0.2 ml of saline at 0, 4 and 20 d. One week after the last sensitization, 20 μ l of OA in saline (40 mg/ml) was administered intranasally, and the drug was administered *p.o.* for 5 d. The next day, histamine (1 ng/ml—10 mg/ml) was administered intranasally at a low dose without drug administration, and the number of sneezes was counted for 10 min. The threshold of the number was determined according to Watanabe *et al.*⁸⁾ using un-sensitized mice. Test drugs were RKS at 30, 100 or 300 mg/kg, ketotifen at 5 mg/kg, seratrodoast at 0.5 mg/kg or 5% arabic gum water (solvent).

Sleeping Time Induced by Pentobarbital in Mice RKS (300 and 2000 mg/kg), ketotifen (5 and 30 mg/kg) or distilled water (solvent) was administered *p.o.* to ICR mice, and 1 h later, 50 mg/kg of pentobarbital was administered i.p. The time between the disappearance and recovery of the righting reflex was measured with a stopwatch.

Gastric Emptying in Rats Rats, fasted for 24 h, received an oral dose of the test drugs, RKS (300 and 600 mg/kg), ketotifen (5 and 10 mg/kg), metoclopramide (3 mg/kg) and distilled water (solvent). Thirty minutes later (RKS, ketotifen and distilled water) or 15 min later (metoclopramide), phenol red at 100 μ g/ml was orally administered, and 15 min thereafter, the animals were sacrificed and the amount of phenol red retained in their stomachs was measured using the method of Yokochi *et al.*⁹⁾

Statistics Each result is shown as the mean $\pm$ S.E. The significance of difference was assessed by the Fisher method for gastric emptying test and the Dunnett method for other examinations.

RESULTS

Antigen-Induced Sneezing in Mice The number of sneezes was counted for 10 min after the antigen challenge to sensitized mice. The number after the topical OA challenge was significantly reduced in the group pretreated with RKS (300 and 1000 mg/kg; *p.o.*) as compared to the group pretreated with distilled water *p.o.* (5.0 ± 1.6 and 2.6 ± 0.5 vs. 16.6 ± 1.8) (Fig. 1).

Antigen-Induced Histamine Release from Rat Peritoneal Exudate Cells The effects of RKS in this test are shown in Fig. 2. The anaphylactic histamine release was not affected by RKS (0.1—1 mg/ml). Tranilast (1 mg/ml) markedly inhibited the anaphylactic histamine release ($p < 0.05$).

Histamine- and Serotonin-Induced Skin Reactions in Rats Table 1 shows the effects of RKS and ketotifen on the exudation of dye in the dorsal skin of rats treated with histamine and serotonin. RKS (300 and 1000 mg/kg) significantly inhibited histamine- and serotonin-induced dye exudation

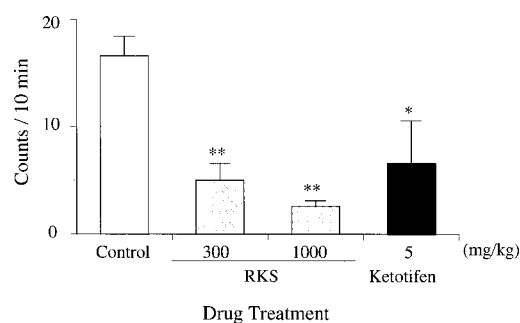


Fig. 1. Effects of RKS and Ketotifen on Antigen-Induced Sneezing in Sensitized Mice

The drugs were administered orally 30 min prior to the antigen challenge. Each column and vertical bar represents the mean $\pm$ S.E. of 5 animals. Significant difference from the control (water 10 ml/kg) at * $p < 0.05$ and ** $p < 0.01$.

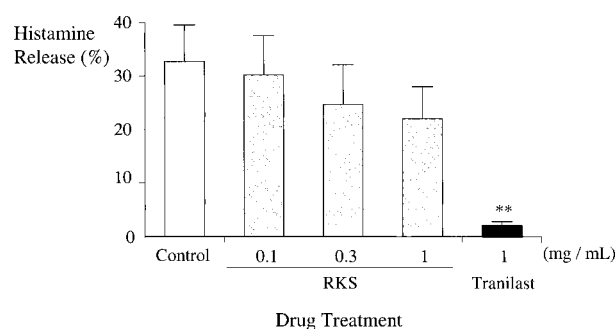


Fig. 2. Effects of RKS and Tranilast on Histamine Release from Rat Peritoneal Exudate Cells

The cells were preincubated for 10 min, then incubated with antigen for an additional 10 min, and the concentration of histamine in the supernatant was determined by HPLC. Each column and vertical bar represents the mean $\pm$ S.E. of 5 experiments. Significant difference from the control at ** $p < 0.01$.

Table 1. Effects of RKS and Ketotifen on Histamine and Serotonin-Induced Skin Reaction in Rats

Drugs	Dose (mg/kg)	Amount of dye leaked (μ g/site)	
		Histamine	Serotonin
Control (water)	5 ml/kg	1.7 ± 0.3	3.9 ± 0.2
RKS	100	1.2 ± 0.3	$2.3 \pm 0.6^{**}$
	300	$0.4 \pm 0.2^{**}$	$1.0 \pm 0.4^{**}$
	1000	$0.1 \pm 0.2^{**}$	$0.2 \pm 0.1^{**}$
Ketotifen	5	$-0.1 \pm 0.1^{**}$	$-0.7 \pm 0.1^{**}$

Each value represents the mean $\pm$ S.E. of 4—5 animals. Drugs were administered *p.o.* 30 min prior to intradermal administration of the allergic mediator (0.1 ml/site). Rats were sacrificed 30 min after administration of the allergic mediator and Evans blue (5 mg/rat, i.v.). Significant difference from the control (water 5 ml/kg) at ** $p < 0.01$.

compared with the control. Ketotifen at 5 mg/kg reduced dye exudation caused by histamine and serotonin.

Histamine-Hypersensitivity in Mice Sneezing was induced by a 1000-times lower concentration of histamine in the control group than in the unsensitized (normal) group (Fig. 3). RKS (100 and 300 mg/kg, *p.o.*) and seratrodoast (0.5 mg/kg) caused a significant improvement in terms of the concentration of histamine that brings about sneezing. Ketotifen (5 mg/kg) had no effect.

Sleeping Induced by Pentobarbital in Mice These effects of RKS are shown in Fig. 4. No effect was observed for

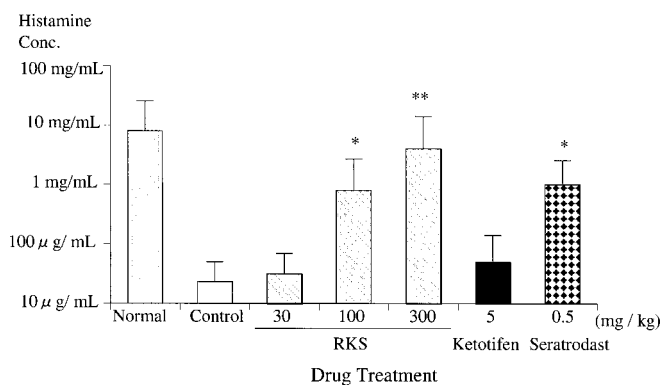


Fig. 3. Effects of RKS, Ketotifen and Seratrodast on Histamine Hyperactivity in Sensitized Mice

Antigen was administered intranasally and test drugs were administered *p.o.* for 5 d in actively sensitized mice. Histamine was given intranasally and the threshold of the sneeze reflex was determined. Each column and vertical bar represents the mean $\pm$ S.E. of 10 animals. Significant difference from the control at * $p < 0.05$ and ** $p < 0.01$.

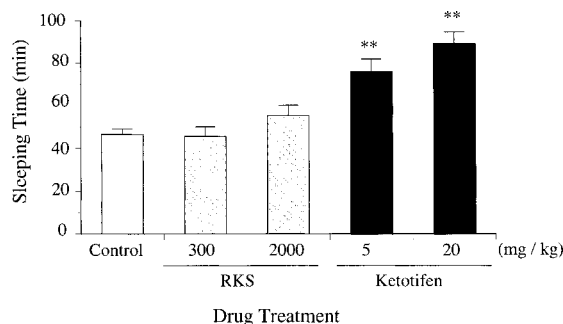


Fig. 4. Effects of RKS and Ketotifen on Pentobarbital-Induced Anesthesia in Mice

Drugs were administered orally 1 h before intraperitoneal injection of pentobarbital sodium. Each column and vertical bar represents the mean $\pm$ S.E. of 10–11 animals. Significant difference from the control at ** $p < 0.01$.

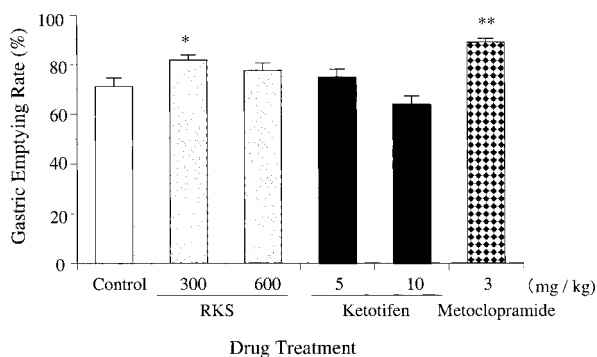


Fig. 5. Effects of RKS, Ketotifen and Metoclopramide on Gastric Emptying in Rats

Drugs were administered orally 30 min (RKS, ketotifen and distilled water) or 15 min (metoclopramide) before phenol red. The amount of phenol red retained in the stomachs was measured 15 min after the administration of dye. Each column and vertical bar represents the mean $\pm$ S.E. of 8–9 animals. Significant difference from the control at * $p < 0.05$ and ** $p < 0.01$.

RKS (300 and 2000 mg/kg), however, ketotifen caused a significant prolongation of sleep at 5 and 20 mg/kg ($p < 0.01$).

Gastric Emptying in Rats RKS (300 mg/kg) as well as metoclopramide (3 mg/kg) significantly accelerated gastric emptying in rats (Fig. 5). RSK (600 mg/kg) and ketotifen (5 mg/kg) had no effect.

DISCUSSION

The drugs used most widely to treat allergic disorders are anti-histaminics, because histamine is considered to play an important role in type I allergies.¹⁰ In this study, RKS ameliorated allergic sneezing and histamine-induced inflammation without inhibiting the release of histamine. Nyunt *et al.* also reported that RKS did not inhibit histamine release from rat peritoneal exudate cells.³ Therefore, we considered that RKS exerts an antiallergic effect, not by inhibiting the release of histamine from mast cells, but by suppressing the response to histamine. Because RKS (0.3 mg/ml and 1 mg/ml) showed an inhibitory effect on the contraction in guinea pig ileum induced by 10^{-6} M histamine (data not shown), RKS may contain histamine H_1 receptor antagonists as active ingredients.

Many antiallergic medicines with antihistaminic activity cause drowsiness.¹¹ It is suspected that this effect is attributable to a blockade of the central histamine H_1 receptor. As shown in Fig. 4, ketotifen, a histamine H_1 receptor antagonist as well as a mast cell stabilizing drug, prolonged sleep induced by pentobarbital in mice. In contrast, RKS showed no effect on pentobarbital-induced sleeping time. Therefore, we want to identify which active ingredients in RKS might lead to the development of blood-brain barrier impermeable histamine H_1 receptor antagonists.

RKS prevented an acceleration of hypersensitivity to histamine in mice. In recent studies, many allergic patients experienced histamine hypersensitivity. The rise in serum IgE levels, eosinophil infiltration, mast cell penetration, phospholipase A_2 (PLA $_2$) and 5-lipoxygenase (5LO) are important factors in the hypersensitive response.^{12–17} PLA $_2$ and 5LO are suggested to be involved in the establishment of hypersensitivity, because a thromboxane A_2 antagonist or a leukotriene D_4 antagonist inhibited airway hypersensitivity.^{18–20} Glycyrrhizin, a triterpenoid saponin contained in a constituent herb of RKS, Glycyrrhizae Radix, selectively inhibits the activities of arachidonate cascade-related enzymes, such as 5LO and PLA $_2$.^{21,22} Therefore, it is possible that glycyrrhizin contributes to the prevention of hypersensitivity by RKS. However, further study will be necessary to identify the active ingredient and the mechanism of action in RKS.

About 1–3% of all patients treated with antiallergic medicines have been reported to develop gastrointestinal disorders. In this study, RKS (300 mg/kg *p.o.*) accelerated gastric emptying in rats, as well as metoclopramide as a positive control. Because RKS showed no effect at the higher dose (600 mg/kg), it remains to be determined whether RKS has beneficial effects on the gastrointestinal system. However, the result suggests that RKS, at least, has no inhibitory effect on gastric emptying.

In conclusion, RKS may be beneficial for the treatment of allergic disorders, especially when patients suffer from drowsiness and gastrointestinal disorders, the most frequent adverse effects of antiallergics.

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